Orthic Andosols

Orthic Andosols are volcanic soils formed from recent volcanic ash, pumice, tephra, or other volcanic parent materials. They are typically dark brown to black, with a very porous and friable structure, low bulk density, and strong aggregation.

The term “Orthic” refers to soils with typical mineralogical properties of Andosols without strong allophanic clay enrichment. These soils develop under humid to sub-humid climates, often on volcanic highlands, and are naturally fertile due to high organic matter content and mineral richness.

Drainage:

- **Wet season:** Well-drained due to high porosity and coarse texture; rapid infiltration prevents waterlogging.
- **Dry season:** Soils dry quickly but remain friable; moisture retention is moderate due to organic matter and volcanic glass content.

Landscape:

Found on volcanic highlands, plateaus, and slopes in regions with historical volcanic activity. Common in central and western Kenya, around Mount Kenya, Aberdare Range, and Rift Valley highlands.

Depth:

Deep (100–200 cm), with rooting generally unrestricted. Shallow areas may occur where ash deposits are thin or eroded.

Workability:

- **When wet:** Soft, loose, and easy to cultivate; tilth is excellent.
- **When dry:** Light and fluffy; minimal crusting but susceptible to wind erosion if left bare.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Moderate (15–25 cmol(+)/kg), depending on clay and organic matter content.
- **Base saturation:** Moderate to high due to volcanic minerals and organic matter.

- **Nutrients:** Nitrogen is often limiting, while phosphorus retention is high due to volcanic minerals; potassium, calcium, and magnesium are generally adequate.
- **Organic matter:** Moderate to high (3–8%), contributing to fertility, structure, and water-holding capacity.
- **Micronutrients:** Iron, manganese, and zinc are usually adequate but pH-sensitive.

pH:
Slightly acidic to neutral (5.5–7.0), depending on rainfall, leaching, and parent material.

Management:

- Apply organic matter (manure, compost, green manure) to maintain fertility and structure.
- Nitrogen fertilization is often necessary to support crop growth.
- Phosphorus fertilizers should be carefully managed due to high phosphate retention.
- Employ erosion control measures on sloping volcanic terrain (terracing, mulching, cover crops).
- Maintain continuous vegetative or mulch cover to prevent wind and water erosion.

Suitable Crops:
High-value crops such as coffee, tea, vegetables (tomatoes, onions, cabbages), root crops (potatoes, carrots), cereals (maize, wheat), and horticultural crops. With proper management, these soils are highly productive under rainfed or irrigated systems.